Simulation Arena

Discount Leakage & Cherry-Picking: The Basket Sabotage

Cherry-pick risk 28/100 (Low): the deep discount here buys nothing on the Hardware line or another offsetting line - in the history we have, that line never went missing on a deal discounted this deeply. The $10k of expected profit at risk (2.1% of the deal) is the ordinary chance of a line dying, not cherry-picking.

Seed 42 | Run run_74bfe9a36daa | 2,000 scenarios tested | Cluster bootstrap resampling with empir… | 21 Sep 2026, 21:07

Same seed, same numbers - this run can be reproduced exactly.

How this works

We went back through your past multi-line deals and found the ones where a line was discounted past 30%. In those deals we measured how often the other lines were cancelled inside 6 months, and compared it with how often those lines die anyway. That difference is the cherry-pick signal. We then rebuilt this quote 2,000 times, each time re-drawing whole past deals so the answer reflects how much history we actually have, and re-priced the basket with whatever survived. The risk score is three things you can check by hand: how far past your discount limit the ask sits (25%), how likely the offsetting lines are to vanish (50%), and how much of the profit sits in the line being cut (25%).

The numbers that matter

CHERRY-PICK RISK SCORE
28 / 100
Built from three checkable pieces; the biggest one is how much of the profit sits in the discounted line.
vs Low risk

PROFIT LIKELY TO LEAK AWAY
$10k
The profit this deal loses once you allow for how buyers actually behave.
vs 2.1% of the deal's profit

CHANCE THE HARDWARE LINE IS DROPPED
0.0%
How often the line that pays for the discount dies in the first 6 months.
vs 0.0x the normal 7.9%

CHANCE ANY OFFSETTING LINE DIES
0.0%
The odds the basket does not survive intact.

PROFIT EXPOSED ON THE HARDWARE LINE
$0
$39k of profit with a 0.0% chance of walking.

PROFIT AT RISK
2.1%
Share of the bundle's profit that is exposed to cherry-picking.

PROFIT IF THEY KEEP EVERYTHING
$489k
The spreadsheet number: 54.3% margin on $900k.

PROFIT YOU SHOULD ACTUALLY PLAN FOR
$479k
The honest number after cherry-picking.
vs as low as $489k in a bad run

How we got there

1. What is actually on the table

The basket is worth $900k and makes $489k if nothing is dropped. But 61.3% of that profit sits in Software - the exact line the buyer wants 35% off, which is 5 points past the discount line you set yourself.

Evidence: $300k of $489k profit in Software

2. What history says happens next

You gave us no history to learn from, so we used your own numbers: 22.0% of lines die anyway, and 64.0% die once a deep discount is given. And they do not do it immediately - when a line dies it happens around month 3.

Evidence: source: your own history

3. What the rebuild shows

Rebuilding this quote 2,000 times, the profit lands at $479k on average, and in an unlucky run of twenty it falls to $489k. There is a 0.0% chance the bundle that survives is worth less than your 35% margin floor.

Evidence: $10k average shortfall

4. What to do about it

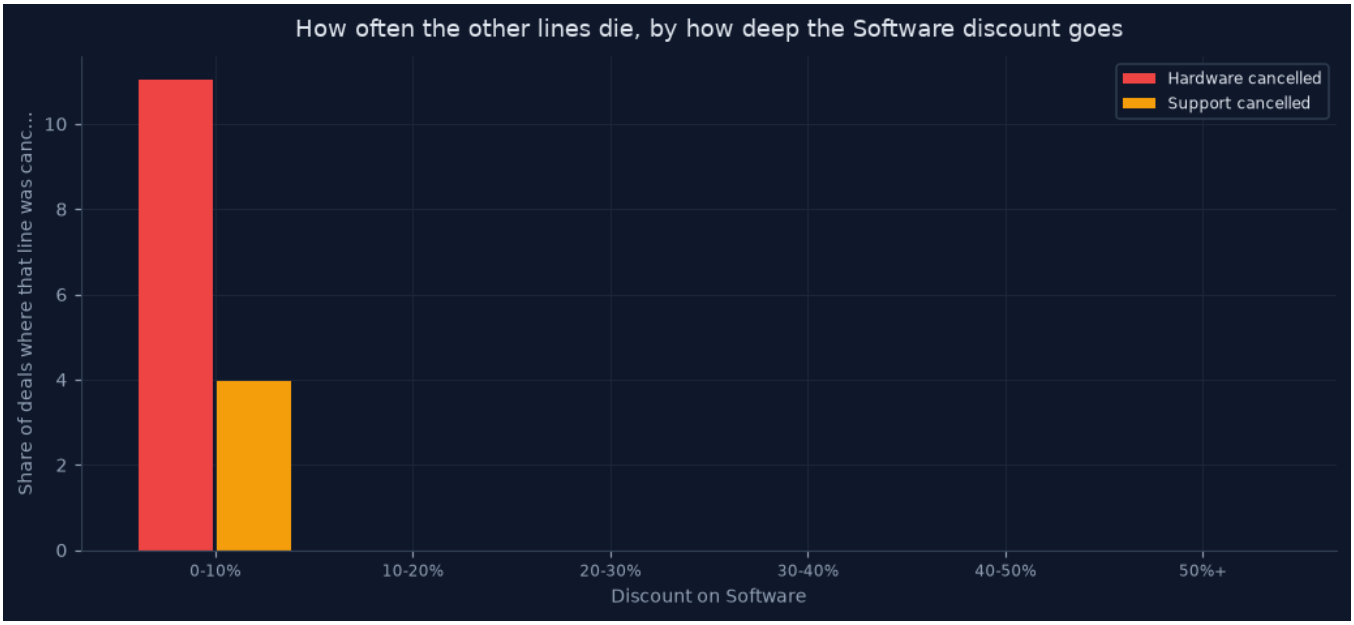
The biggest driver of the score is how much of the profit sits in the discounted line. Tie the discount to the behaviour: if the Hardware line is not live at 80% of committed hours by month 6, the 35% Software discount comes off and Software reverts to standard pricing - worth up to $210k back.

Evidence: clawback up to $210k

Charts

How often the other lines die, by how deep the Software discount goes

Each bar is a group of past deals with a similar discount



What this shows: If the bars jump once the discount passes your limit, buyers are not just buying more - they are buying differently. That jump is the cherry-pick signal.

Which discount poisons which line

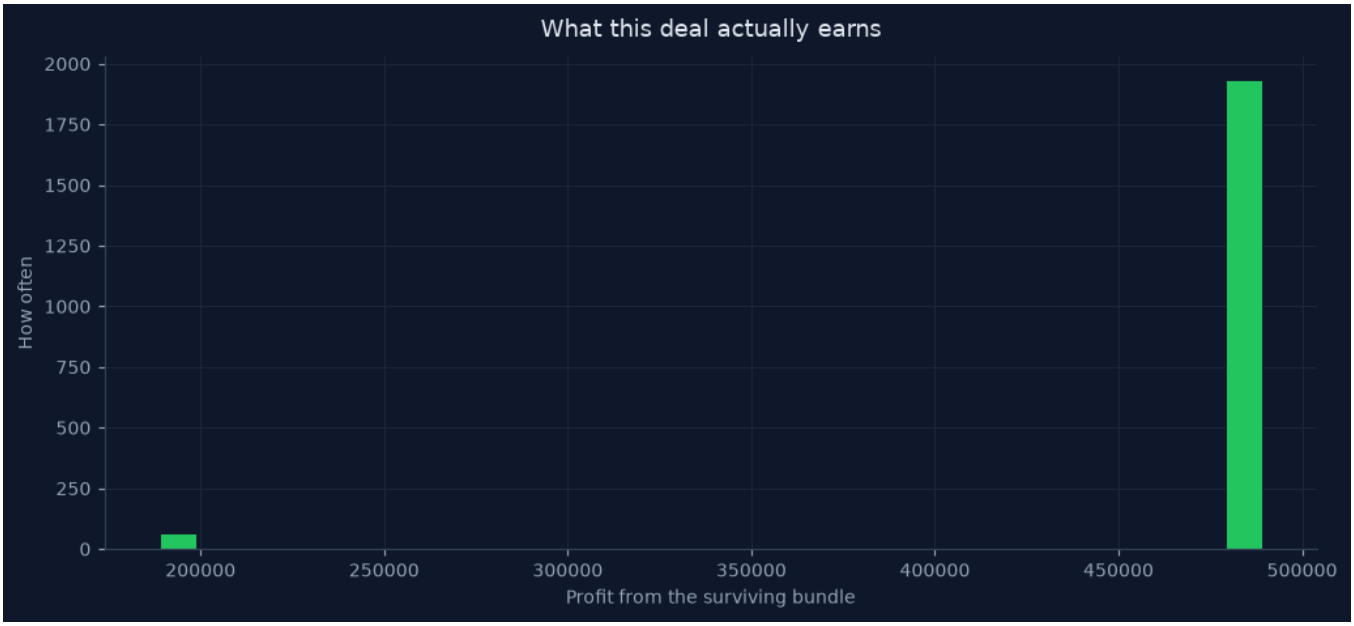
Numbers are 'times more likely than usual' when the row's line is discounted past 30% - rows with too little history are left out



What this shows: Anything above 1.0 means the two events travel together. The hottest cell is the pair you should write a contract clause about.

What this deal actually earns

2,000 rebuilt bundles - the dashed line is the number on the quote



What this shows: The gap between the tall bars and the 'if nothing is dropped' marker is the cost of cherry-picking. The left tail is the bad case you are signing up for.

Tables

The quote, line by line

Where the profit sits, and how exposed each line is.

Line	List price	Discount	Price after discount	Your margin	Profit	Share of profit	Chance it is dropped	Profit at risk
Software	$600,000	35.0%	$390,000	85.0%	$300,000	61.3%	3.6%	$10,729
Hardware	$300,000	5.0%	$285,000	18.0%	$39,000	8.0%	0.0%	$0
Support	$250,000	10.0%	$225,000	70.0%	$150,000	30.7%	0.0%	$0

Row 1 carries the discount being asked for.

What happens to each line when another line is discounted past 30%

Read across a row: 'when THIS line is deeply discounted, the chance THAT line dies'. A figure above 1.0 means the two events travel together.

Discounted line (A)	Line that dies (B)	Deals like this	Normal rate	Rate after the discount	Times more often	Link strength
Hardware	Hardware	251	7.9%	24.7%	3.12x	weak
Hardware	Services	251	16.6%	0.0%	0.00x	weak
Hardware	Software	251	1.0%	0.0%	0.00x	none
Hardware	Support	251	2.8%	0.0%	0.00x	none
Services	Hardware	49	7.9%	0.0%	0.00x	none
Services	Services	49	16.6%	65.3%	3.92x	weak
Services	Software	49	1.0%	0.0%	0.00x	none
Services	Support	49	2.8%	0.0%	0.00x	none
Software	Hardware	2,181	7.9%	0.0%	0.00x	clear
Software	Services	2,181	16.6%	0.0%	0.00x	clear
Software	Software	2,181	1.0%	3.6%	3.70x	weak

Discounted line (A)	Line that dies (B)	Deals like this	Normal rate	Rate after the discount	Times more often	Link strength
Software	Support	2,181	2.8%	0.0%	0.00x	weak
Support	Hardware	11	7.9%	0.0%	0.00x	none
Support	Services	11	16.6%	0.0%	0.00x	none
Support	Software	11	1.0%	0.0%	0.00x	none
Support	Support	11	2.8%	45.5%	15.95x	weak

Rows where the second line dies far more often than normal are the cherry-pick signature.

What the bundle actually earns

Every one of the 2,000 rebuilt bundles, summarised.

Scenario	Profit	Margin	Versus the plan
If they keep everything	$489,000	54.3%	the number on the quote
Most likely outcome	$478,950	53.2%	-$10,050
A bad run (1 in 20)	$489,000	54.3%	$0
Worst bundle we saw	$189,000	21.0%	-$300,000

Row 2 is the number to plan with.

Does the buyer's tier change the behaviour?

Same test, split by customer tier (only tiers with enough history are shown).

Customer tier	Deals	Deals with a deep discount	Chance each offsetting line dies
Enterprise	2,745	583	0%, 0%
Global-2000	1,220	255	0%, 0%
Mid-Market	3,505	777	0%, 0%
SMB	2,461	566	0%, 0%

Use the worst tier as your planning number if you do not know the buyer well.

What we recommend

Approve the 35% only as a conditional discount - expect to lose $10k if you do not tie it to the Hardware line.

Risk Low 28 out of 100 | Confidence: High

Why

• 61.3% of this deal's profit sits in Software, the line being cut by 35%.

• Past deals like this one lose the offsetting lines 0.0% of the time (your own history).

• Expected profit after cherry-picking is $479k, not $489k - a $10k gap.

What to do

1. Make the 35% Software discount conditional: if the Hardware line is not live at 80% of committed hours by month 6, Software pricing reverts to the standard tier (recovering up to $210k).

2. Do not let them drop one offsetting line and keep the other: make the discount conditional on the whole Hardware, Support basket surviving, not on any single line.

3. Put the test at month 6, not at signature - past buyers who walk do it around month 3.

The trade-off: Walking away from the 35% risks the deal; giving it unprotected costs about $10k. The clause lets you keep the deal and cap the loss. If the buyer refuses any condition, the mildest version is to cap Software at 30%, which still protects $30k.

What we assumed and what to watch

Worth knowing before you act

• Some category pairs rest on fewer than 30 past deals, so those figures are rough. The headline pair is the one with enough history behind it.

• Your file holds 14,398,433 rows. We are running on an evenly spread sample of 248,254 of them (1.7%), because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 248,254 rows.

Assumptions

• Quote under review: Software $600,000 at 35% off (85.0% margin), Hardware $300,000 at 5% off (18.0% margin), Support $250,000 at 10% off (70.0% margin).

• Deep discount means anything past 30%; a line counts as abandoned if it dies within 6 months.

• Abandonment rates come from your own history: 9,931 past deals, 2,492 deep-discount category observations.

• Risk score = 25% x how far past your discount limit the ask sits (10 points past = full marks) + 50% x chance an offsetting line dies + 25% x share of profit sitting in the discounted line.

• A cancelled line is treated as fully lost: no revenue and no cost.

• Bundle margin floor 35%; minimum support commitment 80% of hours.

• We replayed 2,000 bundles built from your own past deals; seed 42 - re-running gives identical numbers.

• Quote / deal ID -> your 'quote_id' column

• Customer tier / segment -> your 'customer_tier' column

• Line item ID -> your 'line_id' column

• Product line name -> your 'product_line' column

• Line category -> your 'category' column

• List price of the line -> your 'list_price' column

• Cost of the line -> your 'cost' column

• Discount on the line (%) -> your 'discount_pct' column

• Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column

• Months until cancellation -> your 'months_to_cancel' column

• Data used: 248,254 of the 14,398,433 rows in your file (1.7%), drawn evenly across it.

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
1.20s	prepare	Cost of the line -> your 'cost' column
1.20s	prepare	Discount on the line (%) -> your 'discount_pct' column
1.20s	prepare	Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column
1.20s	prepare	Months until cancellation -> your 'months_to_cancel' column
5.72s	prepare	Read 248,254 records into the model.
5.72s	simulate	Running the simulation…
5.72s	read	Reading the quote you want us to stress-test...
5.73s	read	The basket is worth $900k and carries $489k of profit (54.3% margin) if the buyer keeps every line.
5.73s	read	The line being discounted hardest is Software at 35% off - that is 5 points past your 30% deep-discount line.

Time	Stage	What happened
6.01s	read	Learning from 9,931 past deals across 4 categories (Hardware, Services, Software, Support)...
6.01s	read	Your file holds 248,254 past deals. Rebuilding every one of them thousands of times needs more memory than a browser-class machine has, so the rebuilding works on an evenly spread pool of 9,931 of them (4.0%). Every num…
6.05s	prepare	Checking what happened in past deals that got a discount this deep...
6.05s	prepare	No category pair in your history shows a clear discount-then-cancel pattern - we will fall back on your assumptions.
6.06s	prepare	Chance the offsetting lines disappear: 0.0% - that is what the discount is really buying.
6.06s	simulate	Rebuilding 2,000 bundles that look like this one and re-pricing each of them...
6.39s	simulate	Rebuilt 10% of the bundles...
6.39s	simulate	Rebuilt 20% of the bundles...
6.39s	simulate	Rebuilt 30% of the bundles...
6.39s	simulate	Rebuilt 40% of the bundles...
6.39s	simulate	Rebuilt 60% of the bundles...
6.39s	simulate	Rebuilt 70% of the bundles...
6.39s	simulate	Rebuilt 80% of the bundles...
6.39s	simulate	Rebuilt 90% of the bundles...
6.39s	summarize	All bundles priced. Working out what this deal really earns...
6.39s	summarize	Done. Cherry-pick risk 28/100 (Low) - expect to lose $10k unless the discount is made conditional.